

## Cover Letter

To the Editors of *Symmetry* (MDPI),

I am pleased to submit the manuscript “**Twist Angle Spectrum of Hyperbolic Holonomy: Encoding of Standard Model Flavor Parameters**” for consideration as a research article in *Symmetry*.

**Connection to Symmetry.** The paper is directly concerned with symmetry and asymmetry in the Standard Model flavor sector. The two compact arithmetic hyperbolic 3-manifolds studied—the Meyerhoff manifold  $m003(-2, 3)$  and  $m006(-5, 2)$ —both have first homology  $H_1 = \mathbb{Z}/5$  and are arithmetic, encoding a discrete  $\mathbb{Z}/5$  symmetry. Their invariant trace fields  $\mathbb{Q}(\sqrt{-3})$  (imaginary quadratic) and  $\mathbb{Q}(\sqrt{17})$  (real quadratic) have opposite discriminant signs, and this arithmetic *asymmetry* geometrically explains the lepton/quark CP asymmetry: large leptonic CP violation vs. suppressed quark CP violation at zeroth order. The Iwasawa decomposition  $\mathrm{PSL}(2, \mathbb{C}) = KAN$  distributes the flavor structure symmetrically across mixing (K-factor), CP/mass (A-factor), and Borel overlap (N-factor) channels. The paper is therefore squarely within the journal’s scope: it identifies geometric symmetry structures responsible for observed symmetry-breaking patterns in fundamental physics.

**Summary of main results.** We perform a systematic census of loxodromic twist angles  $\varphi(\gamma) = \mathrm{Im} \log \lambda(\gamma)$  for closed geodesics up to word length 5 on both manifolds and identify numerical correspondences with Standard Model flavor parameters. Three results are especially precise and geometrically motivated:

- The leptonic CP phase:  $\delta_{\mathrm{HFG}} = 195.91^\circ$  (0.55% from PDG 2024), using the word triple fixed independently by the PMNS mixing matrix construction — no free parameters.
- The gauge boson mass ratio:  $M_Z/M_W = 1.1348$  (0.024% from PDG), from a ratio of geodesic lengths on  $m006(-5, 2)$ .
- The reactor angle:  $\theta'_{13} = 8.546^\circ$  (0.07% from PDG), from a single word on  $m006(-5, 2)$ .

The census enrichment factor is  $1.24\times$  ( $p \approx 0.04$ ), honestly presented as modest. We also compute the first Laplace eigenvalue of both manifolds via the Selberg zeta function and propose that the large spectral gap ( $\lambda_1 \sim 2.5\text{--}2.8$ ) geometrically selects SM-scale mixing angles.

**Originality and relation to prior work.** This paper is part of the Hyperbolic Flavor Geometry (HFG) series; companion papers on the CKM matrix (J. Geom. Phys., under review), the CP phase prediction (SSRN 6583551), and the unified framework (SSRN 6775158) are available as preprints. The present paper has not been published or submitted elsewhere. Three factual errors in an earlier preprint version (SSRN 6583549, uploaded April 2026) have been corrected and are explicitly noted in the manuscript with correction remarks.

**Reproducibility.** All computations use SnapPy at 150-bit precision and are fully reproducible via publicly available scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

I confirm that this manuscript has not been submitted to any other journal, that all authors have approved the submission, and that there are no competing interests to declare. The research received no external funding.

I look forward to the possibility of contributing to *Symmetry*.

Sincerely,

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